

QQ Super Compression

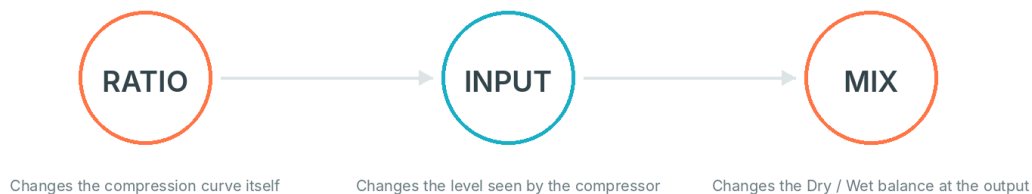
User Manual • English Edition

VERSION 1.0.1

No Attack behavior.

No Release behavior.

Control dynamics while preserving the original attack as much as possible. v1.0.1 adds Threshold, independent LR / MS domains and relative LINK without changing the transparent Lookahead core.



Three ways to control compression - none relies on traditional Attack / Release behavior.

CORE IDEA

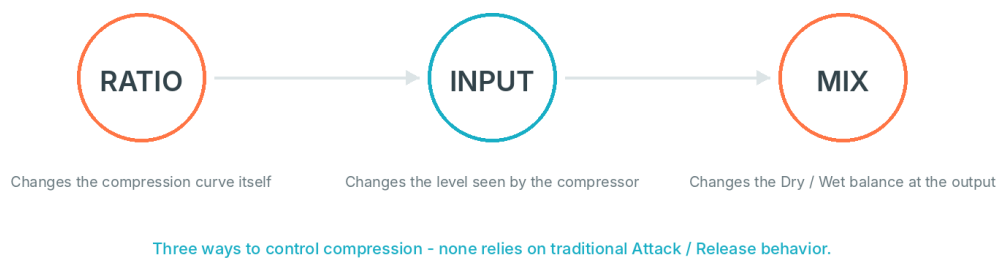
QQ Super Compression still does not start from a traditional Attack / Release compressor model. v1.0.1 adds Threshold, independent LR/MS Ratio / Threshold / Makeup / Mix and relative LINK, while keeping the future-window Peak core. The design accepts a small amount of Lookahead boundary influence in exchange for more stable processing and less carrier-following harmonic coloration.

01 What makes this compressor different is what it does not have

Separate dynamic control from the transient shaping caused by Attack / Release behavior

Traditional compressors do more than reduce dynamics. Their Attack / Release behavior also changes how each note begins and how the gain recovers afterward.

On vocals, guitar, piano, bass and other transient-rich sources, the attack of each note is part of the performance itself. QQ Super Compression is designed for situations where you need dynamic control but want to avoid having traditional Attack / Release behavior reshape those attacks. The key word is behavior - not whether the interface happens to show Attack and Release controls.



v1.0.1 adds Threshold without replacing the compression core

Threshold only defines the level at which the QQ Ratio mapping begins. With Threshold = OFF, the algorithm returns exactly to the threshold-free behavior from before Threshold was added. Enabling Threshold does not add Attack, Release, Knee, an RMS envelope or another Detector.

TRADITIONAL COMPRESSOR

Threshold / Ratio determine how much compression is applied; Attack / Release behavior determines when GR builds and how it recovers, so transient shape changes with those time constants.

QQ Super Compression

A future window first observes the Peak for the corresponding position, then applies the calculated result to the matching delayed waveform. The core has no traditional Attack / Release build-and-recovery process.

TRANSPARENCY FIRST

A longer Lookahead can create a very short pre-influence before an abrupt level change, but testing shows less harmonic coloration than sample-by-sample nonlinear mapping.

0 ms EXCEPTION

At 0 ms, processing becomes much closer to the instantaneous sample domain, so more nonlinear color is intentionally retained and Oversampling is provided.

02 Understand the interface before you start compressing

v1.0.1 enlarges the Dynamic Display and turns Threshold into a practical visual reference



Figure 1 • v1.0.1 main interface - ST / 26 ms / Threshold ON

No.	Area	What it tells you
1	Dynamic Level History	How did the dynamic shape change between the Original, Wet pre-Makeup and final Output?
2	Threshold	At what level does Ratio processing begin, and where does the dashed line sit against the dynamic history?
3	Input / Output / GR Meter	How high are the actual input and final output levels? How much Gain Reduction is produced by the Ratio stage?
4	Input Gain	Do I want the Detector to see a higher or lower input level?
5	Ratio / Makeup / Mix	How should compression strength, compensation and Dry/Wet balance be set?
6	Mode / LINK	How do ST, LR and MS process channels? Should LR/MS preserve the current relationship between both domains?
7	Lookahead / Oversampling	How long is the future observation window? Which Oversampling is used at 0 ms?

03 Threshold: an operating floor, not a different compressor

OFF preserves the original threshold-free QQ algorithm. When enabled, Threshold only decides where compression begins.

Threshold = OFF: every level continues through the original QQ Ratio curve, so the compression core behaves exactly as it did before Threshold was added.

With a numeric Threshold, levels below it remain at 1:1. Above it, the same QQ Ratio curve continues from the threshold point with a continuous transition and no hard step.

WHY THE DISPLAY IS LARGER NOW

Using Threshold is not just about reading a number; it is about seeing where the dynamic history falls above or below the threshold line. v1.0.1 focuses the Display on 0 to -90 dB and shows the Threshold value directly on the dashed line, so you can play the source and find the right boundary visually.

Action	Behavior
Drag Threshold	Continuous adjustment with 0.01 dB parameter resolution.
Shift + drag	Fine adjustment for locating the boundary precisely.
Alt + click	Reset to the default OFF state.
Double-click a value	Enter an exact dB value. Enter OFF or -inf to disable Threshold.
LR / MS	Each domain has its own Threshold and its own dashed line in the corresponding upper / lower Display.

INPUT GAIN AND THE THRESHOLD LINE

The Detector is after Input Gain, while the gray Dry curve shows the Original before Input Gain. To keep the dashed line and the gray reference in the same coordinate system, the Threshold line is repositioned in the Display when Input Gain changes. Its label still shows the Threshold value you actually set.

04 Three ways to control compression: Ratio, Input, Mix

They act at different points in the signal path, but none reintroduces traditional Attack / Release behavior.

Control	What it actually changes	When this approach is useful
Ratio	Changes the strength of the QQ compression curve. With Threshold OFF it acts across the full level range; with Threshold ON the same curve is entered only above the threshold.	The most direct method. Start here when you simply want more or less compression.
Input Gain	Placed before the Detector / Compression stage. Raising Input makes the core see a higher absolute level and also raises the Dry signal inside the processing path.	Natural if you prefer working from input level and gain structure. It also changes the signal position relative to Threshold.
Mix	Placed after Makeup, blending the Input-Gain-adjusted Dry signal with the processed Wet signal.	Useful when you already like a heavily compressed Wet sound and only want to decide how much to blend back in. In LR/MS, Mix is independent for both domains.

WHAT THEY HAVE IN COMMON

Whether you mainly use Ratio, Input or Mix, the QQ core never uses traditional Attack / Release time constants to decide when compression builds or recovers.

THE SAME DOES NOT MEAN MATHEMATICALLY EQUIVALENT

Ratio changes the curve, Input changes the level seen by the Detector, and Mix performs Dry/Wet blending. They cannot be converted into one another exactly. The same only means that none switches to a different Attack / Release envelope.



05 Makeup and Output Gain: loudness compensation and final level

Makeup is before Mix; Output Gain is the final stage of the chain.

Parameter	Position	Role
Makeup	After Compression, before Mix	Compensates the level of the compressed Wet signal. ST uses one control; LR uses separate L/R controls; MS uses separate M/S controls.
Output Gain	Final stage after Mix	Final output trim. It affects the orange Output curve and the Output Meter.
Input Gain	Before Compression	It does more than trim input level; it also changes the level seen by the Detector and therefore the amount of compression.

WHY INPUT GAIN DOES NOT DIRECTLY MOVE THE GRAY DRY CURVE

Input Gain changes the signal that actually enters the compressor and also affects the Input Meter. However, the gray Dry / Input curve in the Dynamic Display always shows the Original before Input Gain. This keeps the reference fixed, making before / after changes easier to see.

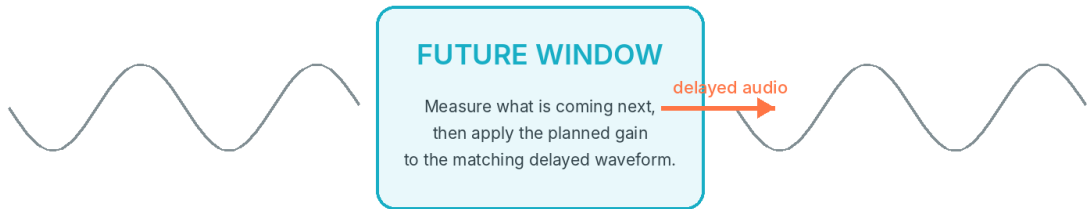
MATCH WRITES TO MAKEUP

Integrated LUFS Match compares Dry with the compressed Wet pre-Makeup and writes the loudness difference once into the Makeup controls for the current mode. ST writes one control; LR writes L/R; MS writes M/S. It is not a continuously running Auto Gain system.

In LR / MS, Makeup, Ratio, Threshold and Mix are fully independent domain parameters; Output Gain remains one shared final output control.

06 Lookahead: it is not Attack

It determines how far ahead the future window can see - not how quickly the compressor clamps down.



QQ Super Compression delays the audible signal while observing peaks in a future window. By the time the matching waveform reaches the output, the processor already knows the level scale inside that future window. It does not need to wait for the transient and then build Gain Reduction through Attack behavior.

Lookahead	Role	Description
0 ms	Special color mode	The Detector falls back to instantaneous sample amplitude, so nonlinearity and harmonics are most obvious.
10 ms	Shorter lookahead	Cleaner than 0 ms while keeping PDC relatively low.
26 ms	Default starting point	A practical balance between low-frequency stability, transparency and latency.
40 ms	More stable	More stable on low frequencies and sustained material.
80 / 100 ms	More transparent	A longer window makes the decision more stable, at the cost of higher total latency.

THE ACCEPTED TIME-BOUNDARY TRADE-OFF

A future window means that if a larger Peak suddenly appears ahead, it can influence a short section immediately before the change. The longer the Lookahead, the longer this microscopic region can become. v1.0.1 deliberately prioritizes this approach because it produces less harmonic coloration and a more transparent result than sample-by-sample nonlinear mapping.

07 0 ms: an intentional sonic option

Oversampling only appears here



Figure 2 • v1.0.1 Lookahead = 0 ms, Oversampling = 8x

0 ms is not the most transparent QQ Super Compression mode. Without a future window to stabilize the carrier cycle, instantaneous nonlinear color becomes much more obvious. Oversampling reduces alias fold-back; it does not turn 0 ms into a transparent mode like 26 ms.

Rate	Why it is included
1x	Preserves the rawest 0 ms sound, with the most aliasing.
8x	Default. In testing, this is the most practical balance between aliasing, CPU use and additional latency.
16x	Further reduces aliasing while retaining the instantaneous nonlinear character of 0 ms.

WHY THERE IS NO 2x / 4x

Earlier PluginDoctor testing showed that 2x / 4x still did not reduce 0 ms aliasing enough, while the extra latency of 8x / 16x remained manageable. The official choices therefore stay at 1x / 8x / 16x. At 10 ms and above the processor is fixed internally at 1x; the Oversampling control is hidden but remembers the last 0 ms choice.

08 ST, LR, MS: one core, three channel relationships

v1.0.1 upgrades LR / MS to fully independent dual-domain processing.

Mode	Compression relationship	Independent parameters	Display
ST	Left and right detect future-window Peaks separately; the stronger result is used for both channels to keep Stereo Linked.	1 × Ratio / Threshold / Makeup / Mix	1 ST Display
LR	Left and Right are detected, compressed and Dry/Wet blended independently.	Separate L/R Ratio / Threshold / Makeup / Mix	Two stacked L / R Displays
MS	The signal is converted to M/S first. Mid and Side are independently detected, compressed and Dry/Wet blended, then decoded back to L/R.	Separate M/S Ratio / Threshold / Makeup / Mix	Two stacked M / S Displays

MODE CONTROL

Mode remains a normal cycling button, not a drop-down menu: ST -> MS -> LR -> ST. Lookahead remains a drop-down menu because it has more preset choices.

WHY LR / MS DISPLAYS ARE STACKED

Using the full horizontal timeline for both domains makes dynamics easier to compare than splitting the display left and right. The upper Display corresponds to L / M, the lower Display to R / S, and the Threshold controls follow the same vertical relationship.

In LR / MS, Input Gain and Output Gain remain shared front-end / final-stage controls. The two processing domains in between are independent.

09 LR: fully independent Left / Right

Left and Right can use different compression strength, thresholds, makeup and Dry/Wet ratios.

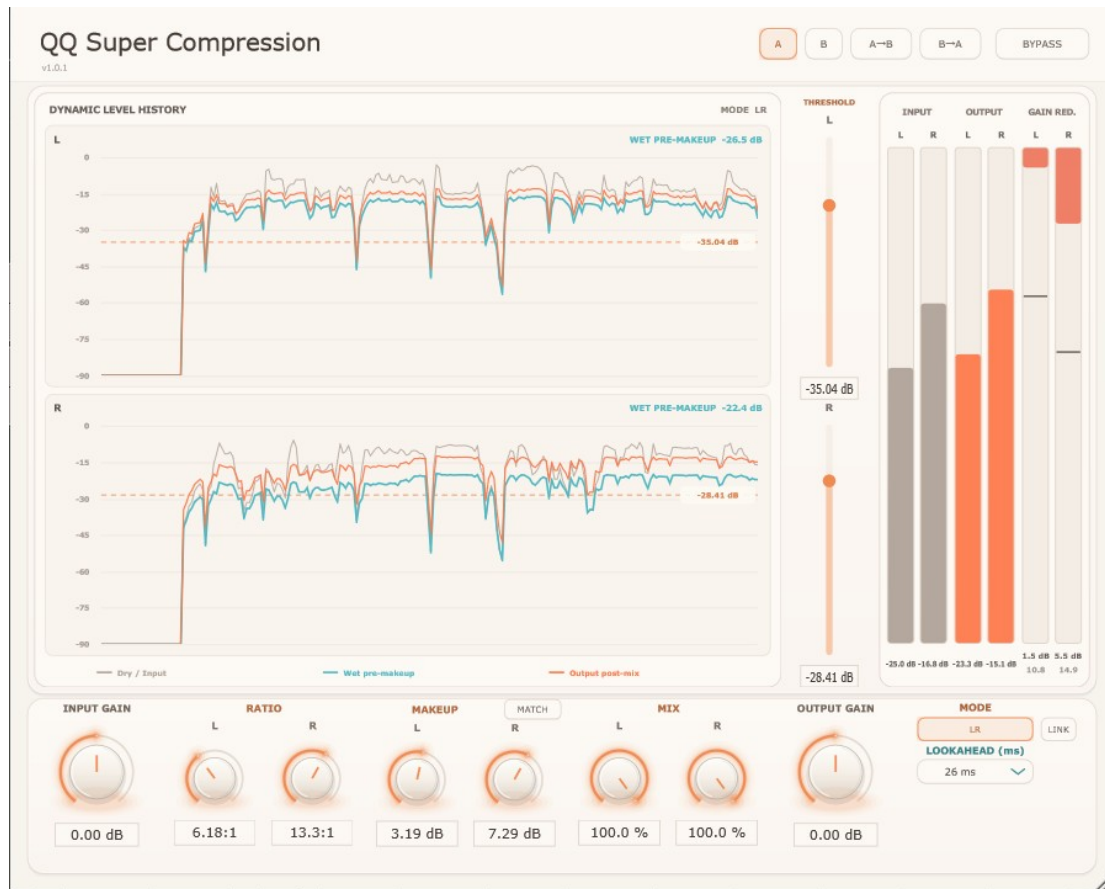


Figure 3 • LR mode: two stacked Displays with separately stored L / R parameters

Independent L / R control

Description

Ratio	Left and Right can use different compression strengths.
Threshold	Left and Right can begin compression at different levels; their dashed lines appear in the upper / lower Displays.
Makeup	Wet loudness can be compensated separately for Left and Right.
Mix	Dry/Wet is independent for Left and Right; for example L 100%, R 70%.

WHEN THIS IS USEFUL

When left/right energy or processing goals are clearly asymmetric, process them independently. If you later want to adjust both together while preserving the difference, turn LINK on.

10 MS: fully independent Mid / Side

Mid and Side do not merely have separate Makeup; they each have independent Ratio / Threshold / Makeup / Mix.

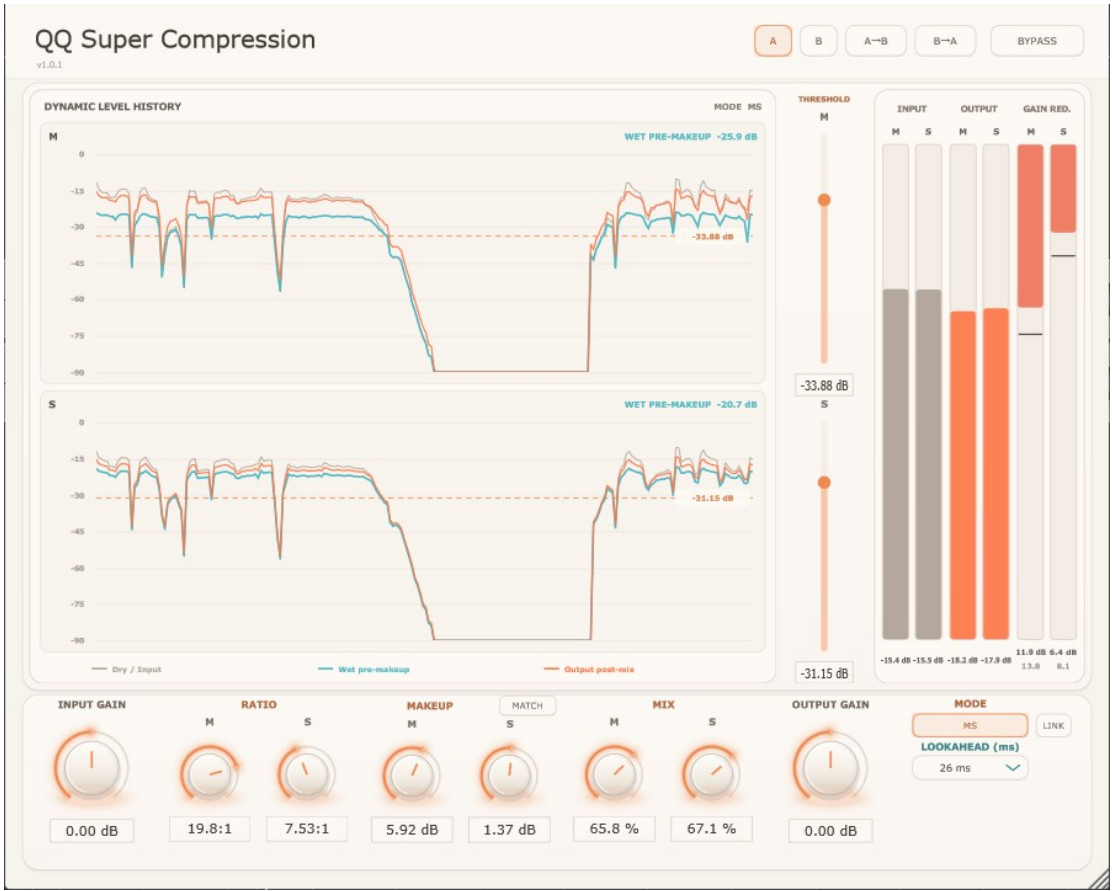


Figure 4 • MS mode: two independent M / S processing domains

Independent M / S control	Description
Ratio	Mid and Side can use different compression strengths.
Threshold	M and S independently decide where Ratio processing begins.
Makeup	M and S use independent Makeup compensation.
Mix	Dry/Wet blending is performed independently in the M/S domain, then decoded back to L/R.

NOTE

MS mode does not simply remix the final Stereo left and right channels. Independent Mix happens inside the M/S domain: M and S each complete their own Dry/Wet blend before being combined back into Stereo.

11 LINK: preserve relative differences, do not force equal values

One button links Ratio, Threshold, Makeup and Mix across the two LR / MS domains.

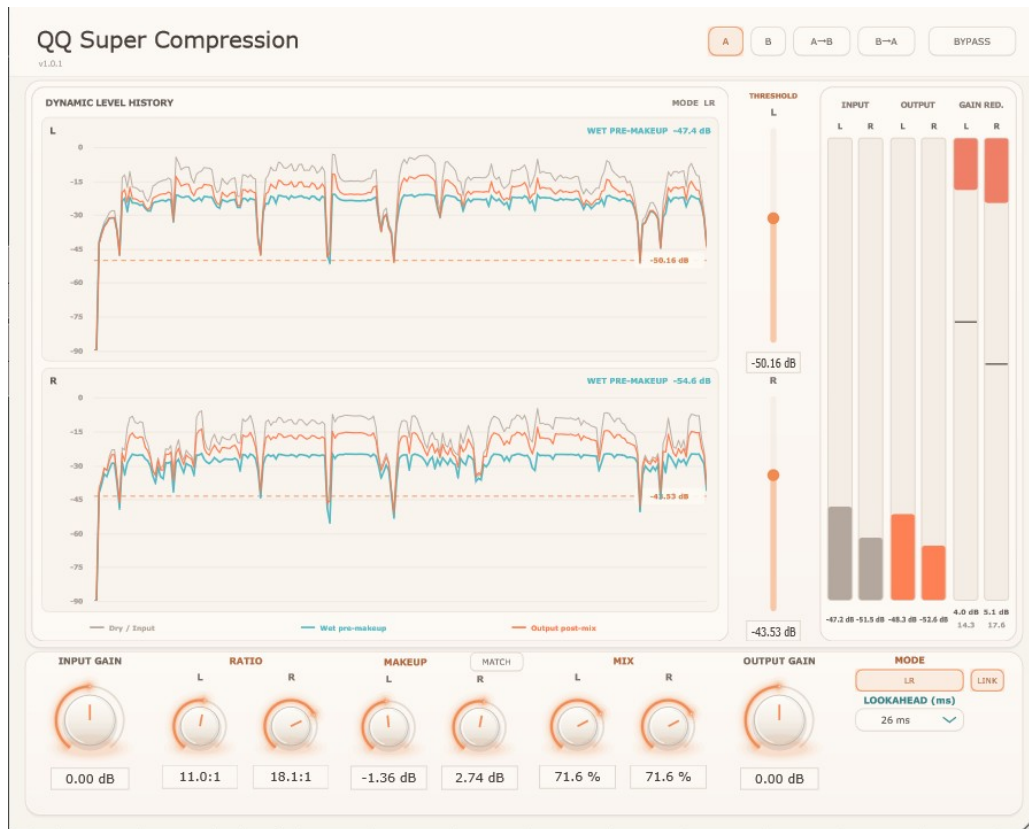


Figure 5 • LR + LINK ON: values remain different while the current relative relationship stays linked

Parameter pair	What LINK ON preserves
Ratio	Preserves the numeric difference. Example: 3:1 / 5:1; first side +1 -> 4:1 / 6:1.
Threshold	Preserves the dB difference. Example: -20 / -10 dB; first side +2 dB -> -18 / -8 dB.
Makeup	Preserves the dB difference. Example: -3 / +1 dB; first side +1 dB -> -2 / +2 dB.
Mix	Preserves the percentage-point difference. Example: 100% / 70%; first side -10% -> 90% / 60%.

NOT SYNCHRONIZE TO THE SAME VALUE

Turning LINK on does not pull both domains to the same values. It preserves the current difference. After that, normal dragging, Shift fine adjustment and direct numeric entry all move the paired parameter by the same amount.

BOUNDARY RULE

Stop at the nearest boundary that preserves the current difference; do not redistribute after a limit is reached. If only one Threshold is OFF, exclude it from finite-dB linking.

12 Dynamic Display: see dynamic changes and find the right Threshold

v1.0.1 fixes the visible range at 0 to -90 dB so useful dynamics occupy more of the display height.



Figure 6 • Dynamic Display + Threshold + Meter work area

Curve / Meter	Position	Meaning
Dry / Input (gray)	Before Input Gain	Original reference signal. The gray line does not move as a whole when Input Gain changes.
Wet pre-Makeup (cyan)	After Compression, before Makeup	Shows the dynamic change caused by the compression core itself.
Output post-mix (orange)	After Makeup + Mix + Output Gain	The final signal sent to the next stage.
Gain Reduction	Ratio stage	0 dB is at the top; deeper downward movement means more GR. Makeup / Mix are not included.

0 TO -90 dB IS ONLY A DISPLAY RANGE

Data below -90 dB is not deleted and does not change DSP, Meter, LUFS or the Threshold parameter. It is simply clamped to the bottom of the Display when drawn, making the useful Threshold range easier to see.

13 Match loudness first, then listen for the difference: Match, A/B and Bypass

If the levels differ before and after compression, it is easy to be fooled by louder sounds better.

Play a representative section and let the Integrated LUFS reading stabilize, then click Match. It writes the loudness difference between Dry and Wet pre-Makeup once into the Makeup controls for the current mode. Then switch Bypass or A/B to hear more clearly what actually changed in dynamics, attack and tone.

Action	Function
Match	One-shot Integrated LUFS matching. ST writes ST Makeup; LR writes L/R Makeup; MS writes M/S Makeup.
A / B	Stores and switches between two sets of sound parameters.
A → B / B → A	Copies the sound parameters from the current side to the other side.
Bypass	Monitors the Original without Input Gain / Compression / Makeup / Mix / Output Gain processing, while preserving the same total PDC.
Double-click a value	Enter an exact parameter value directly. With LINK enabled, paired LR/MS parameters should continue to preserve their relative link.

WHAT A/B STORES

A/B stores Input Gain, the ST/LR/MS Ratio / Threshold / Makeup / Mix values, Output Gain, Mode, Lookahead and the remembered Oversampling choice for 0 ms. LINK is a workflow state saved with the project, but it is not part of the A/B sound snapshot. Bypass is not stored in A/B either.

14 Do not judge by GR alone - return to the source

Decide what you want to preserve before deciding how much to compress.

VOCAL

Start around 26-40 ms. First look at the dynamic distribution in the Display and place Threshold around the phrases / peaks that actually need control. After Match, listen closely to consonants, word attacks and the leading edge of breaths.

PIANO

40-80 ms is often more stable. Listen for whether the hammer attack remains natural while judging whether the overall dynamics sit more easily in the mix.

GUITAR / BASS

Start around 26 ms. Listen for whether pick attack, finger contact, plucks and note leading edges retain their original attack. Do not judge by the GR number alone.

0 ms COLOUR

0 ms + 8x is a good starting point. Use 1x for the rawest color, or 16x when you want to reduce aliasing further.

SUGGESTED WORKFLOW

1) Choose Mode / Lookahead -> 2) Play the source and watch the Dynamic Display -> 3) If you need a threshold, find Threshold first -> 4) Use Ratio / Input / Mix to set the amount of compression -> 5) Use Match / Makeup to align loudness -> 6) Use Bypass / A-B to hear the difference -> 7) Finish with Output Gain to set the final level of the chain.

LR / MS RECOMMENDATION

Start with LINK off and set the two domains to the relative relationship that actually works. If you then want to move both together while preserving that relationship, turn LINK on.

15 Parameter reference and troubleshooting

v1.0.1

Parameter	Range / options	Default / notes
Input Gain	-24.00...+24.00 dB	0 dB; before Compression; shared by all modes.
Ratio	1.00:1...32.0:1	8.0:1; one ST value, separate L/R values in LR, separate M/S values in MS.
Threshold	OFF / -119.99...0.00 dB	Default OFF; 0.01 dB resolution; stored independently for ST/LR/MS.
Makeup	-36.00...+36.00 dB	0 dB; one ST value, separate L/R values in LR, separate M/S values in MS.
Mix	0.0...100.0%	100%; one ST value, separate L/R values in LR, separate M/S values in MS.
Output Gain	-24.00...+24.00 dB	0 dB; final trim; shared by all modes.
Mode	ST / MS / LR	ST; click to cycle.
LINK	OFF / ON	LR/MS workflow link; preserves the relative differences of Ratio / Threshold / Makeup / Mix.
Lookahead	0 / 10 / 26 / 40 / 80 / 100 ms	New instances remember the last choice; the initial starting point is 26 ms.
Oversampling	1x / 8x / 16x	Visible only at 0 ms; default 8x.
Symptom	Explanation / first check	
Input Gain changes, but the gray Dry curve does not	Normal. The gray line always shows the Original before Input Gain.	
The Threshold line moves when Input Gain changes	Normal. The Detector is after Input Gain while the gray Display curve is before Input Gain, so the dashed line is converted into the same reference coordinate system.	
Oversampling only appears at 0 ms	Normal. At 10 ms and above, processing is fixed internally at 1x.	
The DAW briefly realigns when Lookahead changes	The host is updating PDC.	
Match is grayed out	Play valid program material long enough for Integrated LUFS to build a usable reading.	
LR/MS values are different even though LINK is on	Normal. LINK preserves relative differences; it does not force both values to become equal.	
The Display does not show anything below -90 dB	Normal. This is only the drawing range; audio DSP and Meter are unaffected.	

CORE CONCLUSION

QQ Super Compression prioritizes clean, transparent dynamics in real music. Future-window Lookahead is part of that trade-off.

UI Themes - New in v1.0.4

LIGHT

CLASSIC

- Use the LIGHT / CLASSIC button in the upper-right corner to switch the visual theme.
- Both themes have the same layout, controls, parameters, automation and sound.
- Your last selected theme is stored locally and restored the next time the plug-in opens.



Shown above: the CLASSIC theme in QQ Super Compression v1.0.4.